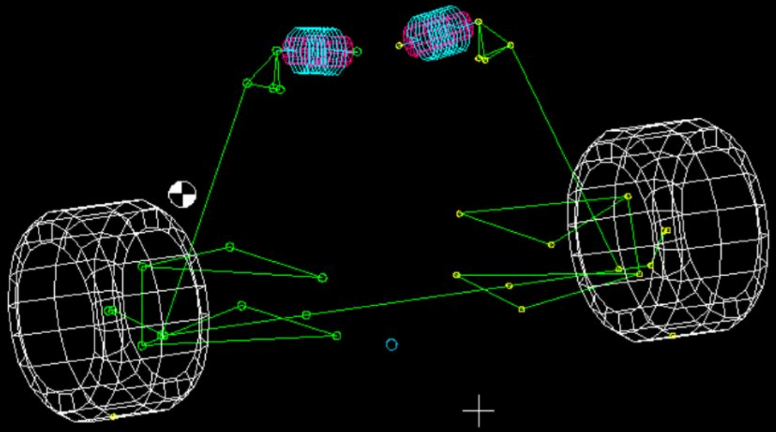
Formula SAE IC | University of Maryland | College Park

Table of Contents

Givens

Geometry

From VD Team
Leader:



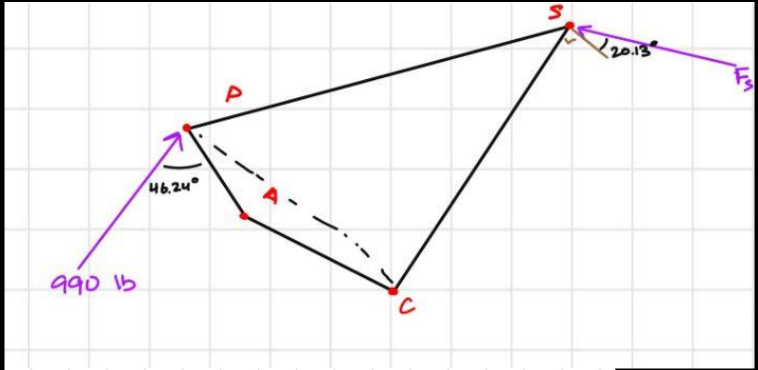
Design Process

FBD and Calculations

Contrary to popular belief, CAD will not always do all the work for you…

Simplify your part to be able to perform manual calculations.

For this part, the force from the spring & damper needed to be calculated.



Angles are between Force and the line between Force and C

F_P : 990 lb $\quad \theta_{PC}$: 75.93°

F_A : 0 lb

F_S : 920.08 lb $\quad \theta_S$: 110.13°

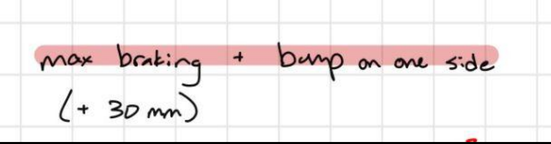
Sum of moments about C:

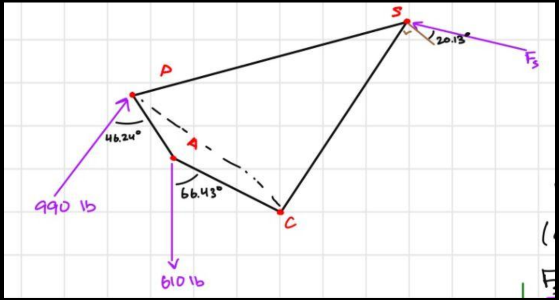
$$F_P \sin\theta_{PC} \cdot \overline{PC} - F_S \cos(20.13) \cdot \overline{SC} = 0$$

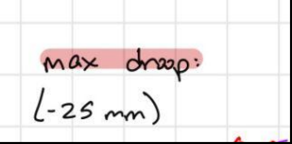
$$(960.30)(66.023102) - 0.9389\, F_S\, (73.393324) = 0$$

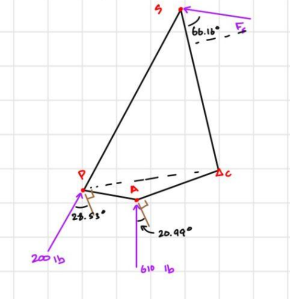
$$F_S : 920.08 \text{ lb}$$

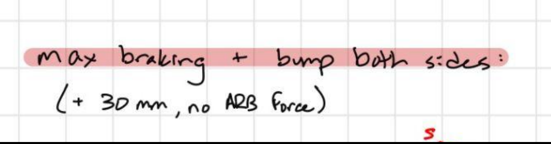
Load Cases

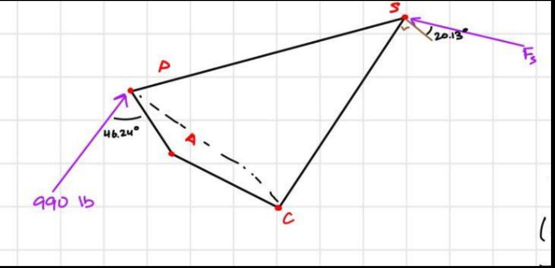

max braking + bump on one side
(+ 30 mm)


S
20.13°
Fs
P
46.24°
A
66.43°
990 lb
610 lb
C


max droop:
(-25 mm)


S
66.16°
Fs
P
A
C
28.53°
20.99°
200 lb
610 lb


max braking + bump both sides:
(+ 30 mm, no ARB force)


S
20.13°
Fs
P
46.24°
A
990 lb
C

Define Goals

- Determine what you're designing for
  - Weight
  - Strength
  - Cost
- Define your constraints
- Prioritize Tradeoffs
  - What is MOST important for the part you're designing?

Objectives:

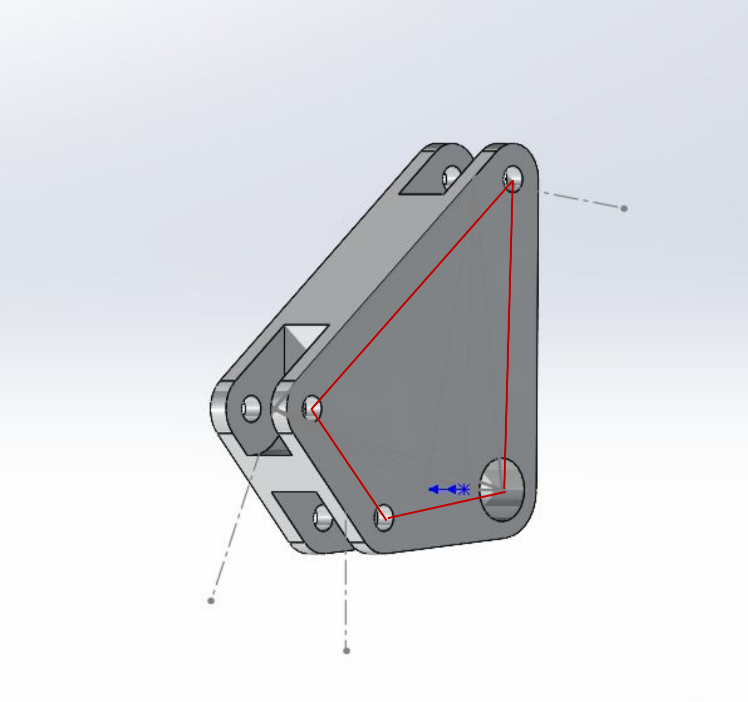
The goal of this part is to transfer and manage forces from the pushrod and the anti-roll bar.

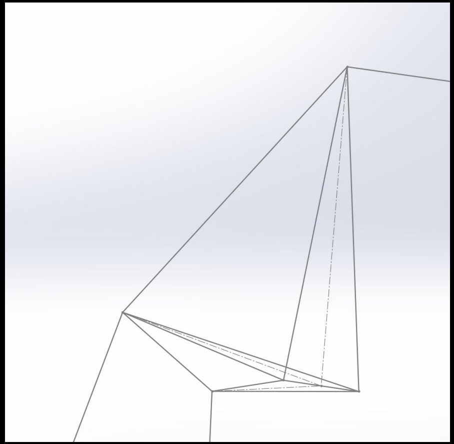
The TR25 front bellcrank should:

- Be able to withstand forces with a safety factor of 2 in three different load cases (note that all of these forces come from Captain's testing data, in which the forces exceed the calculated values):
  - Max Droop: (-)230 lbs from the pushrod, (-)760 lbs from the ARB
  - Max Braking and Bump on One Side: (-)1440 lbs from the pushrod and 760 lbs from the ARB
  - Max Braking and Bump on Both Sides: (-)1440 lbs from the pushrod
- Have a motion ratio – the ratio of wheel travel to spring travel – of around 0.85
- Provide clearance for other suspension parts (rod ends, spring and damper, bolt heads, etc.), ensuring that there is no contact during rotation
- Fit within the established TR25 suspension geometry
- Utilize bearings that can withstand radial forces with minimal friction during rotation
- Be as lightweight as possible
  - My personal goal is a 15% reduction from last year's FBC
- Be machinable with CNC Machine
- Cost efficient

Block Design

Given geometry shown with red lines:





Material Selection

Material Selection

Criteria:	Weight:	Aluminum 7075-T6	Aluminum 6061-T6	4130 Steel	Titanium (Grade 5)
Weight/Density:	5%	4	5	1	3
Availability:	5%	5	5	5	2
Cost:	10%	4	5	4	2
Stiffness (Elastic Modulus):	15%	3	1	4	5
Fatigue Resistance:	15%	5	3	5	5
Machinability:	20%	4	5	2	1
Strength-Weight Ratio:	30%	4	3	2	5
Total:	100%	4.05	3.5	3.05	3.65

Final Choice: Aluminum 7075-T6

Even if you KNOW what you will be using, justify it.
This is helpful for explaining to judges at comp.

Bearing Selection

Criteria:	Weight:	Needle Roller:	Oil Embedded Bushing:	Ball Bearing:	Nadella Combination
Sealing:	5%	4	2	3	3
Durability:	10%	2	5	2	2
Cost:	10%	4	5	2	1
Low Friction Coefficient:	15%	5	1	4	5
Size/Weight:	20%	3	5	3	2
Radial Load Capacity:	40%	5	4	2	5
Total:	100%	~~4.15~~	3.85	2.55	3.6

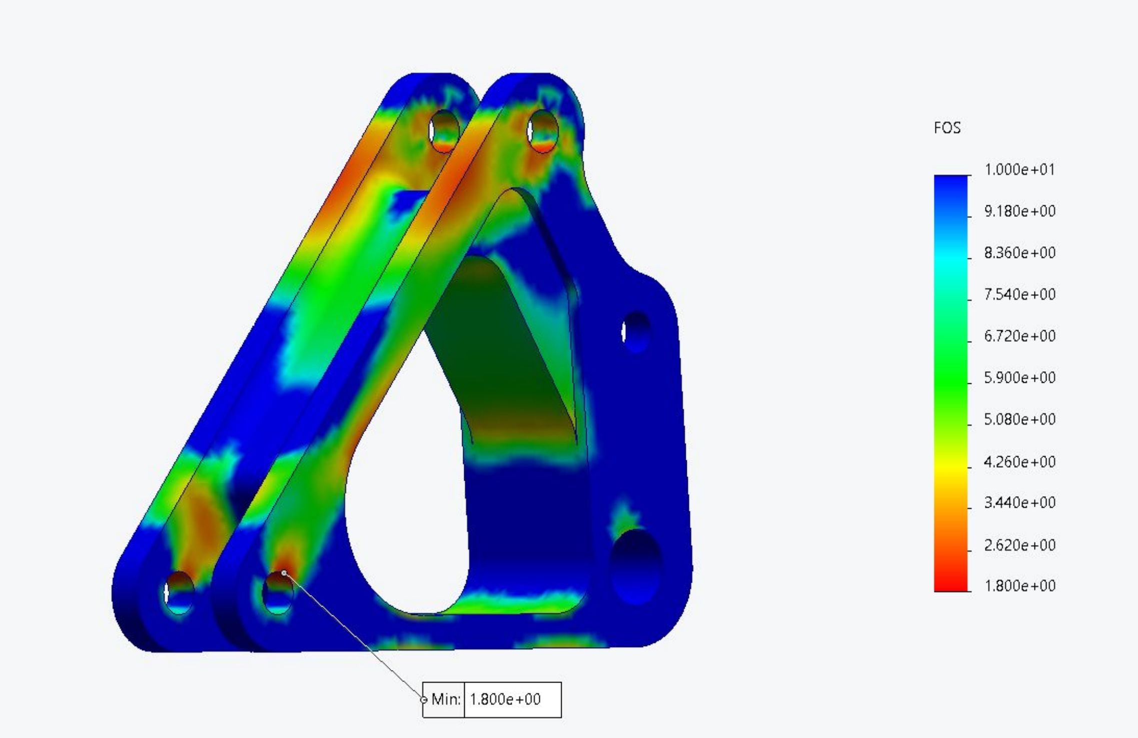
Final Choice: Oil Embedded Bushing (https://www.mcmaster.com/2938T735/)

Not everything works out as you'd expect…

The bearing that looked "best" had too much play so we had to pivot…

Simulation and Lightweighting

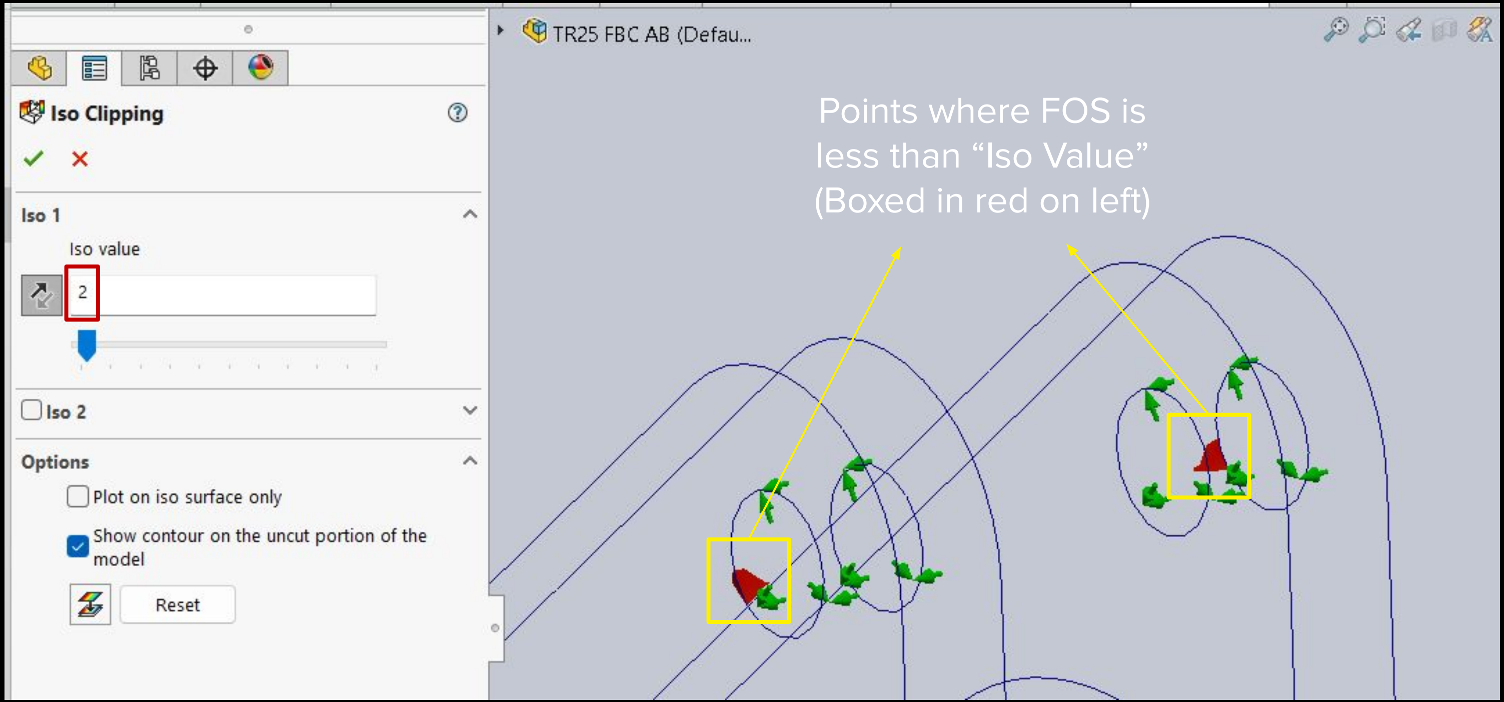
FEA


FOS
1.000e+01
9.180e+00
8.360e+00
7.540e+00
6.720e+00
5.900e+00
5.080e+00
4.260e+00
3.440e+00
2.620e+00
1.800e+00
Min: 1.800e+00

Where to Start

- Experiment
  - In this case, topology optimization was attempted. It wasn't used but gave ideas for where to go
- Start simple
  - Start with a basic design
  - Develop many iterations
  - Brainstorm with others, fresh eyes are typically helpful

FEA (Iso Clipping)

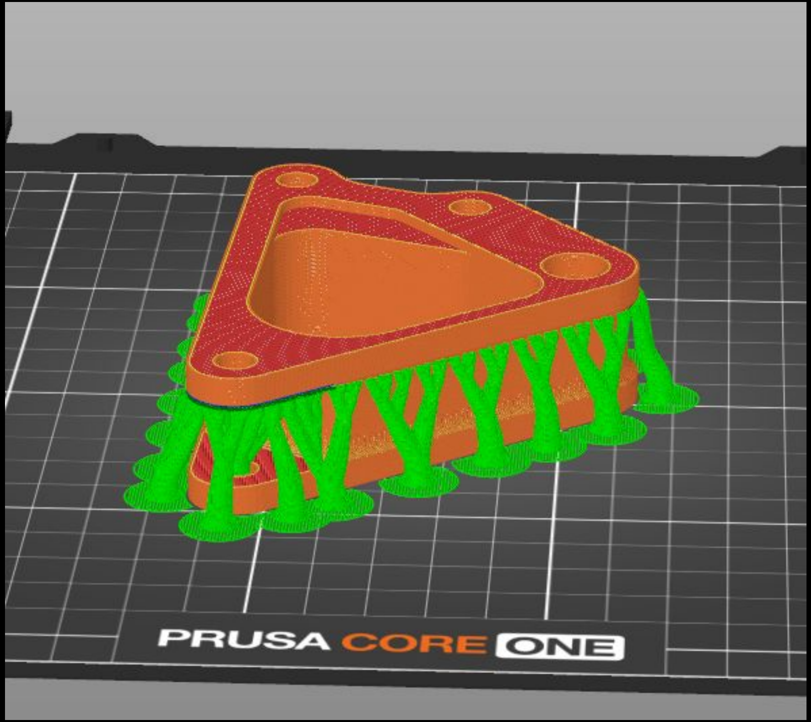


Prototyping & Manufacturing

Prototype (Typically 3D Print)

This is helpful for many reasons:

- Test bolt / bearing clearance
- Check that part fits in car (clearance with other parts, fits in mount…)
- Spot faults in design more easily
    - It is much easier to notice problems on a physical part than a model on a computer

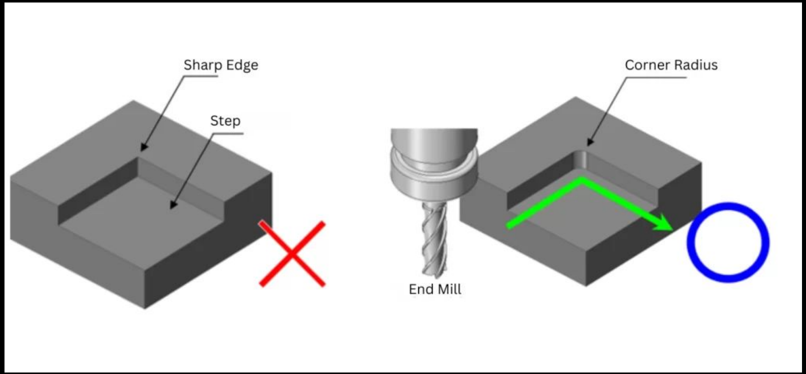


Design for Manufacturing

Typical Design for Manufacturing (DFM)

Principles:

- Keep it simple, avoid any unnecessary fancy shapes
- Use fillets, not sharp corners (also helpful for reducing stress!)
- Avoid super thin walls



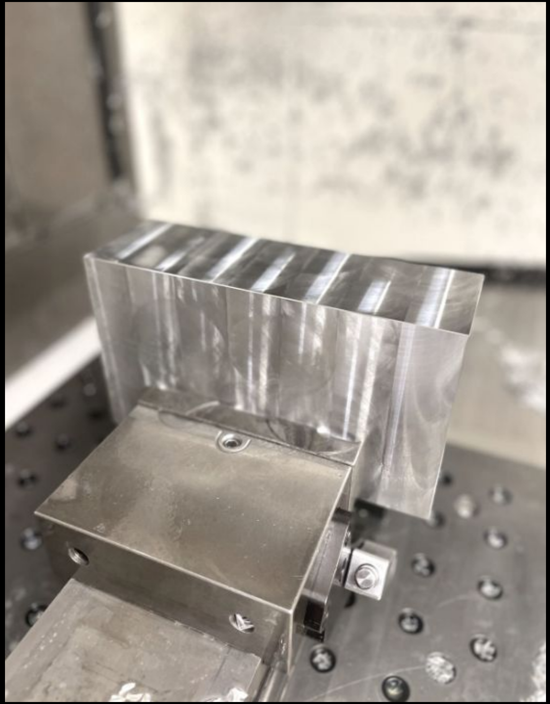
Design for Manufacturing (cont.)

Watch videos on design for manufacturing (DFM) for the manufacturing method you plan on using.

If you have any questions, ask whoever will be manufacturing your part. They should know the limits of that manufacturing process very well.

This will be a separate presentation.

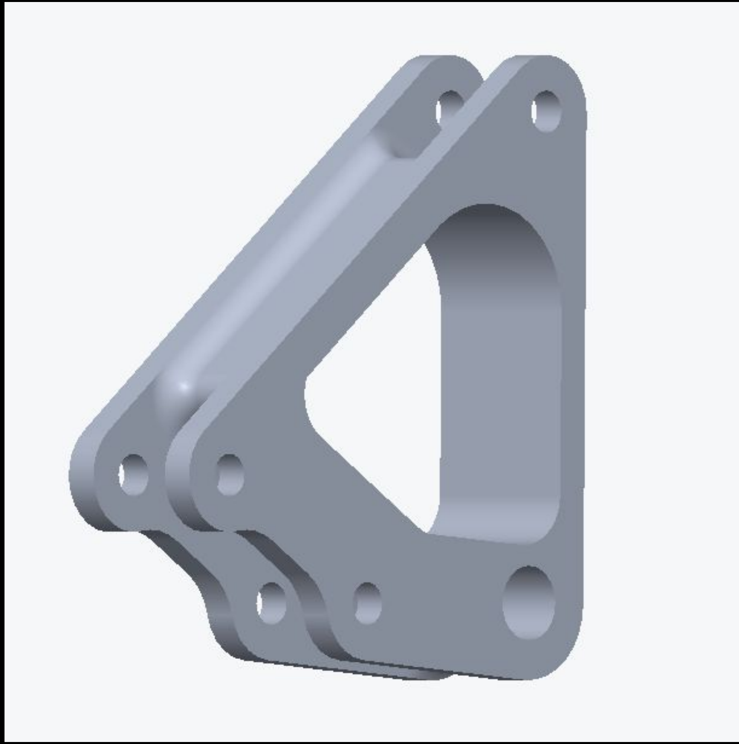
This part was made with CNC Machining.



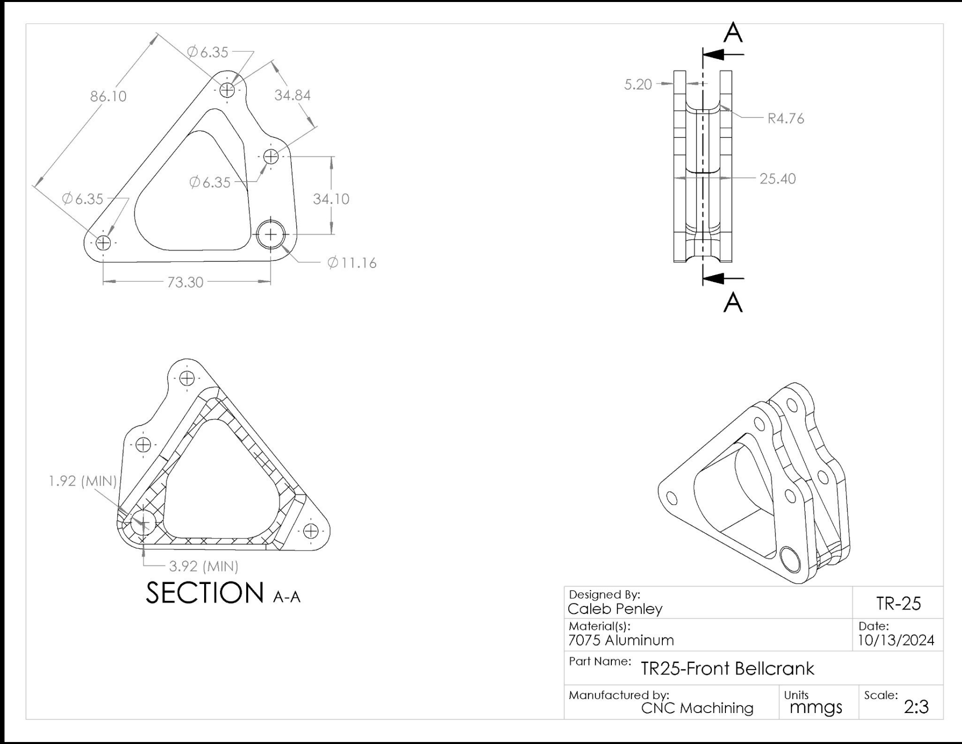
https://youtu.be/qx_qqVmjCc0?si=uJyCNkCKuyOGTxcC

Final Product

Final Design



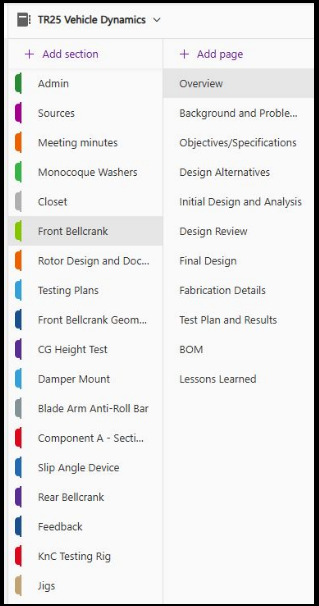
Drawing

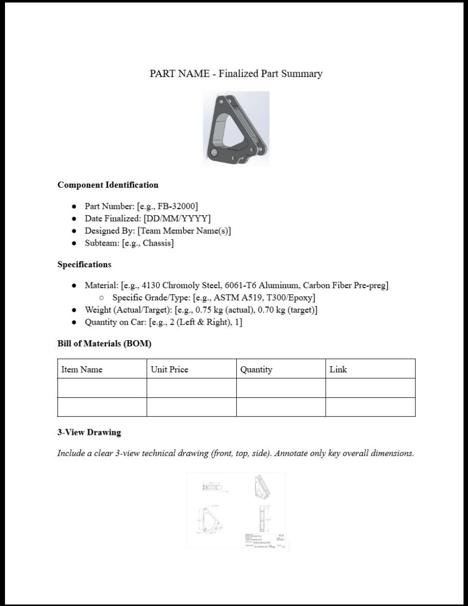


Ø6.35
86.10
34.84
Ø6.35
Ø6.35
34.10
Ø11.16
73.30
A
5.20
R4.76
25.40
A
1.92 (MIN)
3.92 (MIN)
SECTION A-A
Designed By:
Caleb Penley
TR-25
Material(s):
7075 Aluminum
Date:
10/13/2024
Part Name:
TR25-Front Bellcrank
Manufactured by:
CNC Machining
Units
mmgs
Scale:
2:3

Document, document, document

- Document all steps in OneNote
- **Final parts must be sent to Subteam lead and Chief Engineer before implementation**
- Fill out [Finalized Part Summary]()
    - FSAE 2026 -> Admin -> Templates





PART NAME - Finalized Part Summary

Component Identification

- Part Number: [e.g., FB-32000]
- Date Finalized: [DD/MM/YYYY]
- Designed By: [Team Member Name(s)]
- Subteam: [e.g., Chassis]

Specifications

- Material: [e.g., 4130 Chromoly Steel, 6061-T6 Aluminum, Carbon Fiber Pre-preg]
    - Specific Grade/Type: [e.g., ASTM A519, T300/Epoxy]
- Weight (Actual/Target): [e.g., 0.75 kg (actual), 0.70 kg (target)]
- Quantity on Car: [e.g., 2 (Left & Right), 1]

Bill of Materials (BOM)

Item Name	Unit Price	Quantity	Link

3-View Drawing

Include a clear 3-view technical drawing (front, top, side). Annotate only key overall dimensions.

Helpful Tips

Check yourself before you wreck yourself

Make sure you:

- Review FSAE rules to be absolutely positive you're compliant
- Run FEA correct units (N vs. lbf, this mistake was made...)
- Check load cases and calculations
- Ensure final FEA is run with high fidelity mesh
- Confirm part can be assembled, bolts can go in and out...